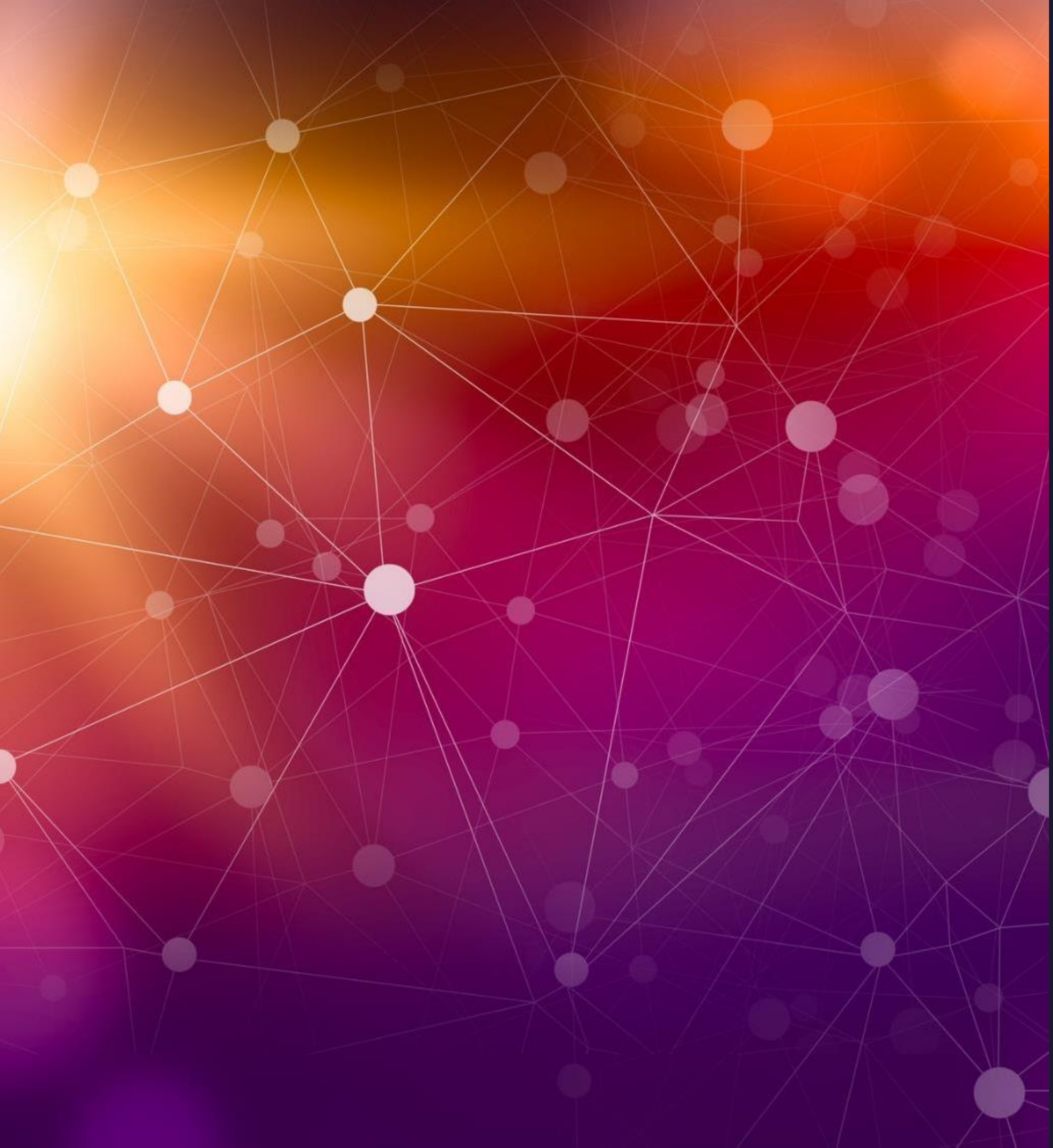

# Day in the life of Lead Analytics & AI

*- Mohd Amaluddin Yusoff*

# Agenda

Brief Bio, Career Timeline

A day at work or A week at work

Projects

- Digital Twin, Predictive Analytics, Enterprise ChatGPT

Q&A


# Mohd Amaluddin Yusoff

- My current role is Lead Analytics and AI in an oil and gas (O&G) industry.
- Hold a PhD, research areas was on the applications of mathematical optimization in engineering.
- Before joined O&G industry, I was a Senior Lecturer with Curtin University and an Electronics Engineer developing transmission control software for Caterpillar, Inc. in the USA.
- I am passionate in Artificial Intelligence, teaching and programming. Therefore, I am currently leading a team to apply machine learning algorithms at work, and coaches secondary school students as part of company's Social Investment program.

# Career Timeline

**EMBEDDED ENGINEER**

Caterpillar, Inc. in the USA.


**LECTURER**

Electrical and Electronics Engineering, Digital System.

**OIL & GAS**

Cost Estimator, Electrical Engineer.

**DATA SCIENCE**

Data Scientist,
Lead Analytics & AI


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# My typical activities as a Lead Analytics and AI

Business engagements include solution discovery, requirements gathering, stakeholder ~~update meeting~~, etc.

Coaching team members especially on machine learning or predictive analytics

Capability building of Analytics community of practice

Leading adoption of digital or AI tool

Deliver digital initiatives

Building the team

# Projects

# Digital Twin

- ❑ Virtual representation of a physical item: Elements, structure, behavior
  - ✓ Contextual data visualization and interaction
  - ✓ Past, Present, Future information - adaptive and predictive.
- ❑ Foundational for digital transformation and data strategy
  - ✓ Drives use cases, shows what data is needed, sparks collaboration & ideas.
- ❑ Enablers
  - ✓ Data integration including Internet of Things (IOT)
  - ✓ Advanced Visualization (3D)
  - ✓ Data Synthesis (Advanced Analytics/ML)

**Aggregate**

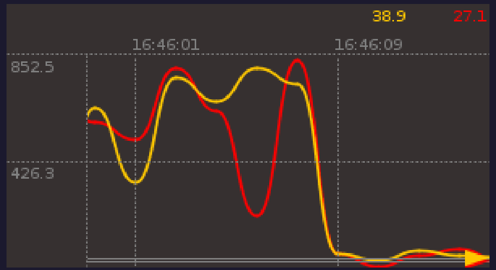
- Integration of Data from Multiple data sources & data formats
- Secure but easy access & consumption

**Analyze & Enhance**

- Transform data into contextual information & insight
- Derive value from data (Analytics/AI)

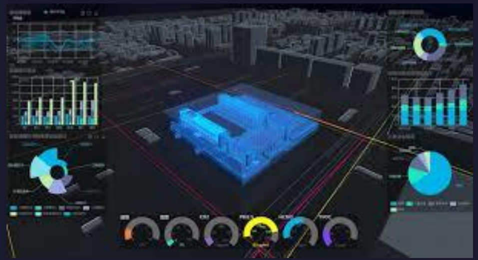
**Visualize & Act**

- Data in easily understandable format (2D/3D)
- Past , current, future information




Predictive Analytics Solution

Information Management Solution



Interactive Dashboards

# Asset 3D

“Your plant at your fingertips”

- Virtual representation of the physical assets and equipment on-site in three dimension.

- **Capabilities/ Benefits**
  - Provides users with highly realistic and detailed visualization of assets, making it easier to understand their features, characteristics and components
  - Allows for better job scoping and handover discussions during maintenance and turnaround activities
  - Can be used for training and educational purposes for new hires, allowing users to interact with the plant in a realistic and immersive way

# Asset 3D

“Your plant at your fingertips”

# Asset 3D

"Your plant at your fingertips"

# Predictive Analytics for Maintenance

Why use Machine Learning for Predictive Maintenance?

- ❑ More integrated – not single disciplines
- ❑ Automated monitoring
- ❑ It is possible identify hidden patterns
  e.g., composite interactions of many sensors

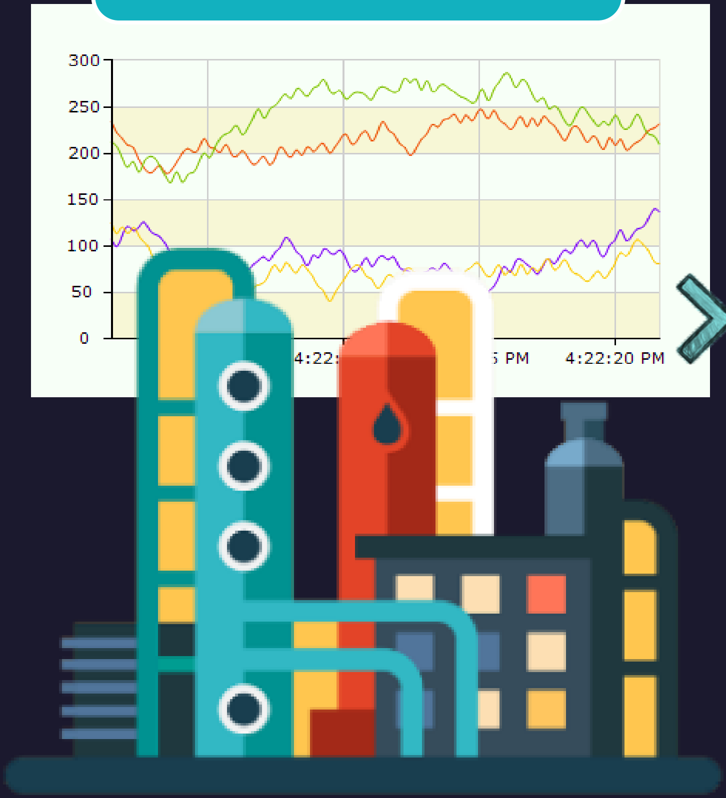
# Predictive Analytics

# Predict Asset Failure

**1**

### Historical Data

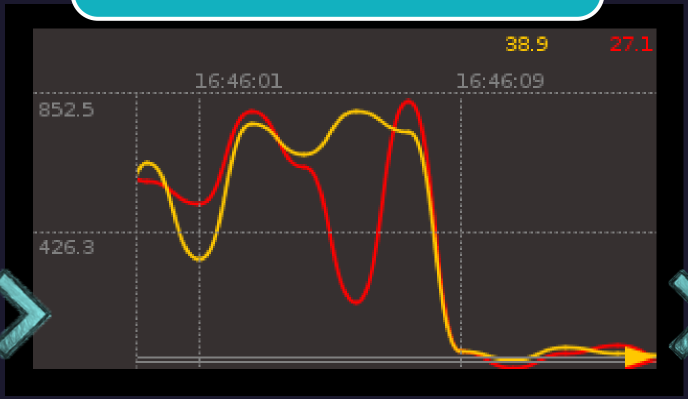
Application learns normal operation from historical data




**2**

### Pattern Recognition

Advanced algorithms create and organize operational profiles




**3**

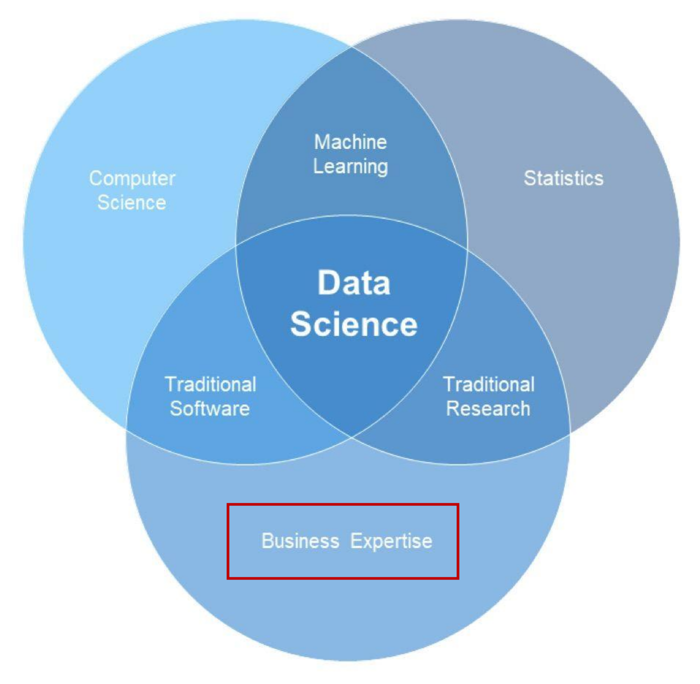
### Early Warning

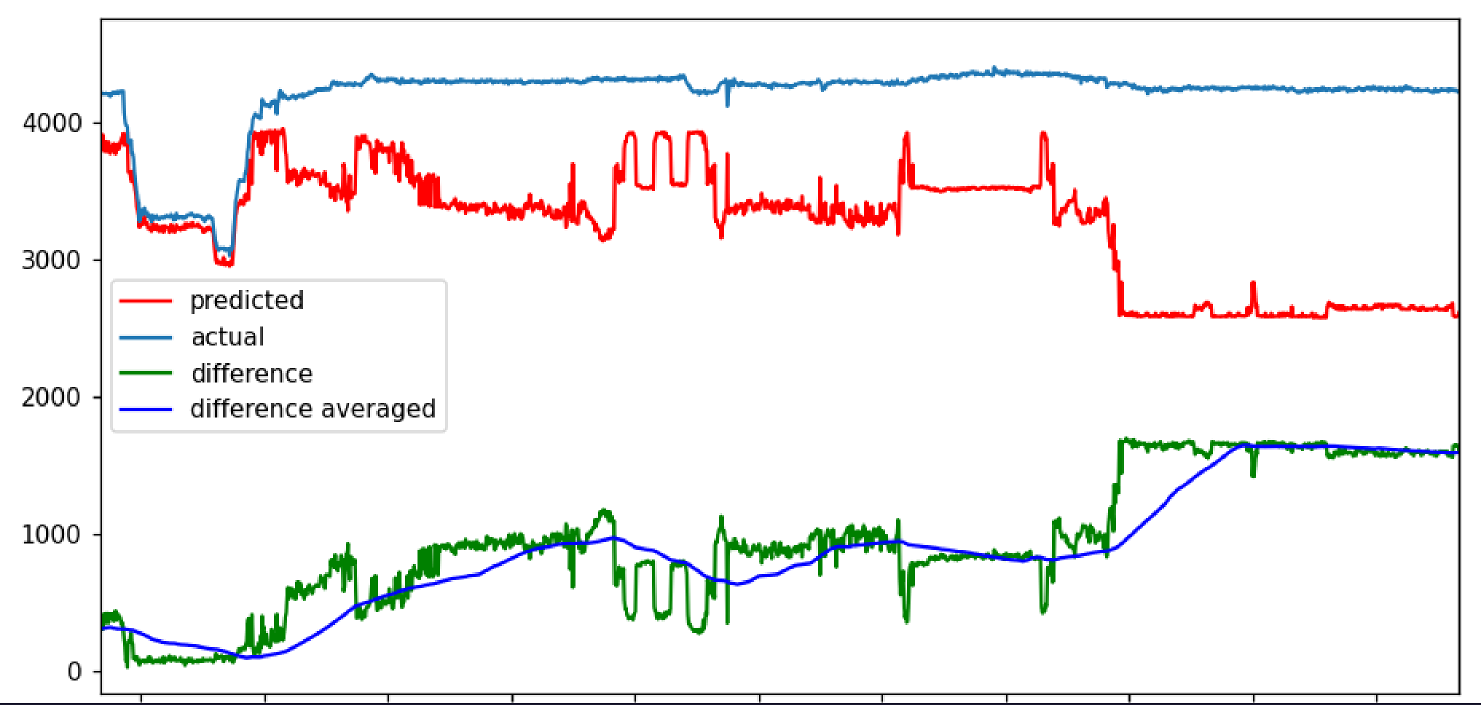
Deviations from normal operation are identified and displayed

- Provides machine learning and predictive capabilities which helps in early identification of equipment health and performance problems
- **Capability**:
  - Predictive analytics to enable Asset integrity and Production Performance improvement
  - Application learns normal operation from historical data
  - Advanced algorithms create and organize operational profiles
  - Deviations from normal operation are identified and displayed providing early warning

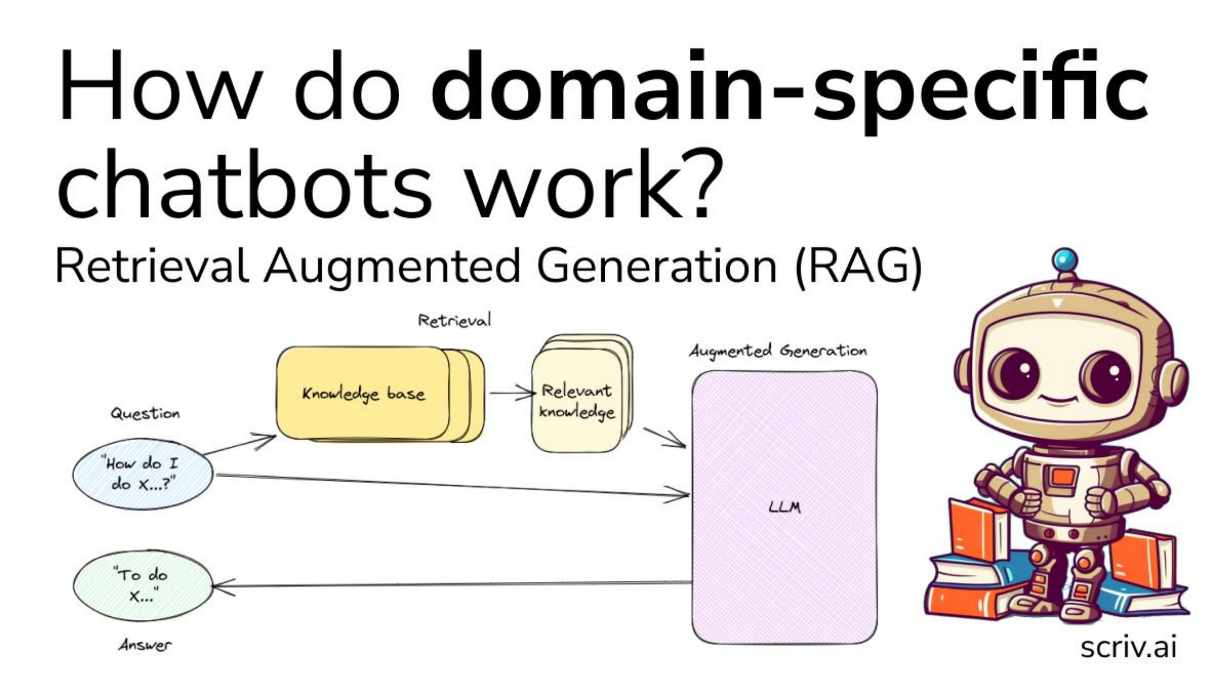
# Regression for Valve Analytics

- To predict flow rate using historical data
- If predicted flow rate differs 'too much' compared to actual flow rate, then something is wrong!



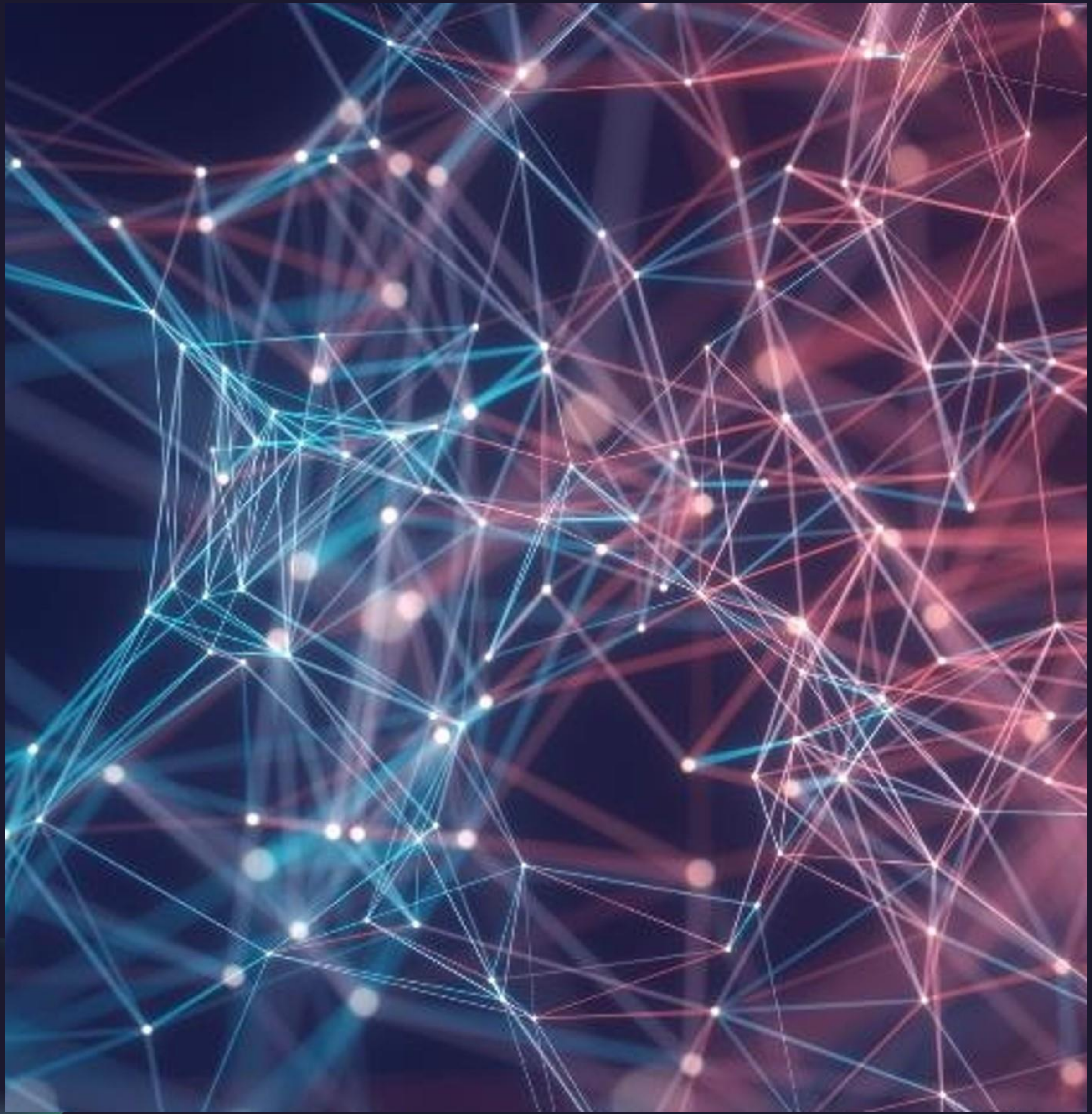




# Enterprise ChatGPT




- RAG-based chatbots provide a natural language interface that enhances employee productivity.
- An enterprise knowledge base also allows for the chatbot to have access to company specific information.
- LLM combined with an enterprise knowledge base is very valuable. Why?

# Q&A



Thank you